

we introduce the PG-VarMiON emulating the optimal Petrov-Galerkin formulation, describe the training procedure, and present an analysis of the generalization error. Numerical results are presented in Section 4 for the diffusion and advection-diffusion equations in one dimension, and the advection-diffusion problem in two dimensions. We end with concluding remarks in Section 5.

2 Problem Formulation

Let $\Omega \in \mathbb{R}^d$ be an open, bounded domain with piecewise smooth boundary Γ . The boundary is further split into the Dirichlet boundary Γ_D and natural boundary Γ_η , with $\Gamma = \Gamma_D \cup \Gamma_\eta$. Define the space $H_D^r(\Omega) = \{u \in H^r(\Omega) : u|_{\Gamma_D} = 0\}$. We consider the following scalar elliptic boundary value problem

$$\begin{aligned}\mathcal{L}(u(\mathbf{x}); \mathbf{g}(\mathbf{x})) &= f(\mathbf{x}) & \forall \mathbf{x} \in \Omega, \\ \mathcal{B}(u(\mathbf{x}); \mathbf{g}(\mathbf{x})) &= \eta(\mathbf{x}) & \forall \mathbf{x} \in \Gamma_\eta, \\ u(\mathbf{x}) &= 0 & \forall \mathbf{x} \in \Gamma_D,\end{aligned}\tag{2.1}$$

where $\mathcal{L}$ is a linear elliptic PDE operator and $\mathcal{B}$ is the natural boundary operator, both parametrized by a set of functions $\mathbf{g} \in \mathcal{G}$. Also, $f \in \mathcal{F} \subseteq L^2(\Omega)$ is the source term, and $\eta \in \mathcal{H} \subseteq L^2(\Gamma_\eta)$. The solution $u \in \mathcal{V} := H_D^r(\Omega)$, where r depends on the order of the operator $\mathcal{L}$.

A particular example of (2.1) is the steady advection-diffusion equation with

$$\begin{aligned}-\nabla \cdot (\kappa(\mathbf{x}) \nabla u(\mathbf{x})) + \mathbf{c}(\mathbf{x}) \cdot \nabla u(\mathbf{x}) &= f(\mathbf{x}) & \forall \mathbf{x} \in \Omega, \\ \kappa(\mathbf{x}) \nabla u(\mathbf{x}) \cdot \mathbf{n} &= \eta(\mathbf{x}) & \forall \mathbf{x} \in \Gamma_\eta, \\ u(\mathbf{x}) &= 0 & \forall \mathbf{x} \in \Gamma_D,\end{aligned}\tag{2.2}$$

where $\mathcal{V} = H_D^1(\Omega)$, $\mathbf{n}$ is the unit outward normal on Γ_η and the set of parametrizing functions are $\mathbf{g} = [\kappa, \mathbf{c}]$. Here $\kappa \in L^\infty(\Omega) \cup \{\kappa \mid \kappa(\mathbf{x}) \geq \kappa_{\min} \text{ a.e. } \mathbf{x} \in \Omega\}$ for some (fixed) scalar $\kappa_{\min} > 0$ is the diffusion coefficient, while $\mathbf{c} \in H_{\text{div}}^1(\Omega) = \{\mathbf{c} \in [L^2(\Omega)]^2 \mid \nabla \cdot \mathbf{c} \in L^2(\Omega)\}$ is the velocity field. We will use (2.2) as a canonical example for the numerical results in Section 4.

2.1 Variational form and symmetrization

The variational formulation of (2.1) is given by: find $u \in \mathcal{V}$ such that

$$a(u, w; \mathbf{g}) = (f, w) + (\eta, w)_{\Gamma_\eta} \quad \forall w \in \mathcal{V},\tag{2.3}$$

where $(\cdot, \cdot)$ is the $L^2(\Omega)$ inner-product, $(\cdot, \cdot)_{\Gamma_\eta}$ is the $L^2(\Gamma_\eta)$ inner-product, while $a(u, w; \mathbf{g})$ is the associated bilinear form parameterized by $\mathbf{g}$. We also assume that the bilinear form is coercive, which requires additional conditions on $\mathbf{g}$. With this assumption, a unique solution of (2.3) exists, as guaranteed by the Lax-Milgram theorem [10, 9].

For the particular case of the advection-diffusion equation, we have

$$a(u, w; \kappa, \mathbf{c}) := (\kappa \nabla u, \nabla w) + (\mathbf{c} \cdot \nabla u, w)\tag{2.4}$$